PHYS 143 – Mechanics and Thermal Physics

Motion

Reference and practice deck

Supporting material cut from the live lecture. Nothing here is new physics.

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How to use this deck

Labels

REFERENCE keep for later use: definitions, equation anatomy, toolkits, derived results.

PRACTICE use to develop the skill: concept checks, extra examples, independent problems.

Where things are

Part A Contact 1: skills list, velocity anatomy, calculus and graph toolkits, extra examples.
Part B Contact 2: acceleration and constant-a anatomy, toolkits, extra free-fall problems.
Part C Contact 3: projectile, circular and relative-velocity anatomy and toolkits, curved paths, parametric equations, extra problems.
Part D the chain of ideas and the seven toolkits.

The live lecture is `lecture02_motion.pdf`; the master toolkit there is the one procedure to use first.

Slide titles that mention a Toolkit refer to the seven toolkits listed in Part D.

What to master, and what supports it

Core skills you should master

- distinguish the six quantities of motion;
- choose an origin and positive direction;
- differentiate $\vec{\boldsymbol{r}}(t)$ to find $\vec{\boldsymbol{v}}$ and $\vec{\boldsymbol{a}}$;
- integrate $a(t)$ with initial conditions;
- read and sketch x–t, v–t, a–t graphs;
- decide from the signs of v_x, a_x if speed rises;
- choose a constant-a equation, say when it applies;
- solve free fall, choosing the physical root;
- solve projectiles with components linked by time;
- explain and calculate radial acceleration;
- add velocities with the subscript chain.

Supporting tools

- the parametric form of a trajectory;
- tangential and normal acceleration on a general curve;
- derivations and the calculus refresher (supplement).

Supporting slides carry the label *supporting tool* in their titles. They stay in the deck for reference; the core skills are what the problems test first.

Extra example: a dragster (Young Ex. 2.1)

Question

A dragster is at $x_1 = 19\,\mathrm{m}$ at $t_1 = 1.0\,\mathrm{s}$ and at $x_2 = 277\,\mathrm{m}$ at $t_2 = 4.0\,\mathrm{s}$. Find its average x-velocity.

Predict first

Positive, and roughly what size? (It covers about $250\,\mathrm{m}$ in about $3\,\mathrm{s}$.)

Extra example: a dragster (Young Ex. 2.1)

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Predict first

Positive, and roughly what size? (It covers about $250\,\mathrm{m}$ in about $3\,\mathrm{s}$.)

Identify: target $v_{\mathrm{av},x}$; known two positions and times. **Set up:** x along the track, positive forward.

Execute: $v_{\mathrm{av},x} = \dfrac{277 - 19}{4.0 - 1.0}\,\mathrm{m/s} = \dfrac{258}{3.0} = 86\,\mathrm{m/s}.$

Evaluate: positive, units m/s, consistent with the estimate ($\approx 83\,\mathrm{m/s}$). ✓

Cf. Young and Freedman, Example 2.1.

But what if... the dragster was slower than $86\,\mathrm{m/s}$ at the start and faster at the end? What was its velocity at exactly $t = 2.0\,\mathrm{s}$? → **instantaneous velocity**.

Concept check: distance or displacement?

PRACTICE

Which of these can be negative? Which is path-independent?

1. The length of road travelled from Accra to Kumasi.
2. The change in position from Accra to Kumasi, with x measured north from Accra.
3. The change in position of a lap runner after one complete lap.
4. The odometer reading minus the starting reading.

Concept check: distance or displacement?

Which of these can be negative? Which is path-independent?

1. The length of road travelled from Accra to Kumasi.
2. The change in position from Accra to Kumasi, with x measured north from Accra.
3. The change in position of a lap runner after one complete lap.
4. The odometer reading minus the starting reading.

Answers: (1) distance, never negative. (2) displacement, positive here. (3) displacement $= 0$, distance $=$ one lap. (4) distance. Displacement depends only on the end points; distance depends on the whole path.

But what if... we measure x westward instead? Every sign in the runner's journey flips ($\Delta x = -60\,\mathrm{m}$), yet nothing physical changed. We need a habit that fixes the signs first: **Toolkit 1**.

TOOLKIT 1 – Signs and coordinates

WHEN TO USE: any problem with a position, displacement, velocity or acceleration along a line; every later toolkit this week starts here.

PROCEDURE: **Choose** an origin and a positive direction *before* assigning any sign, and mark them on the sketch. **Assign** signs relative to *your* axis: $v_x > 0$ means moving toward $+x$, $v_x < 0$ toward $-x$; the speed is $|v_x|$. A change in anything is final minus initial.

WATCH OUT: reversing the axis flips every sign; the physics does not change.

CHECK: does each sign agree with the sketch and with the motion described?

But what if... we now ask how rapidly the runner's position changed over the $30\,\mathrm{s}$? That needs one number for the whole journey: **average velocity**.

Equation anatomy: velocity

EQUATION – average velocity

$$v_{\text{av},x} = \frac{\Delta x}{\Delta t} = \frac{x_2 - x_1}{t_2 - t_1}$$

Meaning. the net change of position per unit time over an interval.

Variables. Δx is final minus initial position; Δt is the elapsed time.

Units. $\mathrm{m/s}$.

Interpretation. the sign gives the direction of the net motion; it is not the average speed.

Example. the dragster, $+86\,\mathrm{m/s}$; the runner, $+2.0\,\mathrm{m/s}$.

EQUATION – instantaneous velocity

$$v_x = \lim_{\Delta t \to 0} \frac{\Delta x}{\Delta t} = \frac{\mathrm{d}x}{\mathrm{d}t}$$

Meaning. the velocity at one instant: the limit of the average over ever shorter intervals.

Variables. $x(t)$ is the position as a function of time.

Units. $\mathrm{m/s}$.

Interpretation. the slope of the tangent to the x–t graph; its sign is the direction, its size is the speed.

Example. $x = 2t^2$: $v_x = 4t = 12\,\mathrm{m/s}$ at $t = 3\,\mathrm{s}$, as the table of chords suggested.

TOOLKIT 2 – The calculus of motion

WHEN TO USE: the position (or velocity) is given as formulas in t and you need the velocity (or acceleration). The reverse step, integration, is added in Contact 2.

$$\textbf{position } \vec{\boldsymbol{r}}(t) \xrightarrow{\mathrm{d}/\mathrm{d}t} \textbf{velocity } \vec{\boldsymbol{v}}(t) \xrightarrow{\mathrm{d}/\mathrm{d}t} \textbf{acceleration } \vec{\boldsymbol{a}}(t)$$

PROCEDURE: **Differentiate component by component:** $\dfrac{\mathrm{d}}{\mathrm{d}t}t^n = nt^{n-1}$, $\dfrac{\mathrm{d}}{\mathrm{d}t}\cos\omega t = -\omega\sin\omega t$, $\dfrac{\mathrm{d}}{\mathrm{d}t}\sin\omega t = \omega\cos\omega t$. **Substitute** the time *after* differentiating.

WATCH OUT: the argument of $\sin$ and $\cos$ is in radians; ωt is dimensionless.

CHECK: every derivative divides by seconds, so the units must change accordingly; the direction of $\vec{\boldsymbol{v}}$ matches the sketch of the path.

Independent problem

A particle has $\vec{r} = (3.0t^2)\,\hat{\imath} + (4.0t)\,\hat{\jmath}$ (SI units). Find its velocity at $t = 2.0\,\mathrm{s}$, its speed, and the direction of motion.

Predict first

Sketch first. Is the particle moving mainly in x or y at $t = 2\,\mathrm{s}$? Speeding up or slowing down in x?

Independent problem

A particle has $\vec{\boldsymbol{r}} = (3.0t^2)\,\hat{\boldsymbol{\imath}} + (4.0t)\,\hat{\boldsymbol{\jmath}}$ (SI units). Find its velocity at $t = 2.0\,\mathrm{s}$, its speed, and the direction of motion.

Predict first

Sketch first. Is the particle moving mainly in x or y at $t = 2\,\mathrm{s}$? Speeding up or slowing down in x?

$\vec{\boldsymbol{v}} = 6.0t\,\hat{\boldsymbol{\imath}} + 4.0\,\hat{\boldsymbol{\jmath}} \;\Rightarrow\; \vec{\boldsymbol{v}}(2) = (12\,\hat{\boldsymbol{\imath}} + 4.0\,\hat{\boldsymbol{\jmath}})\,\mathrm{m/s}. \quad v = \sqrt{144 + 16} = 12.6\,\mathrm{m/s}.$
$\theta = \tan^{-1}(4.0/12) = 18.4^\circ$ above $+x$ (quadrant I).
Evaluate: mainly x, as predicted; units m/s. ✓

But what if... we are given only a *graph* of $x(t)$, with no formula to differentiate? Then slopes must be read off the picture: **Toolkit 3**.

TOOLKIT 3 – Reading motion graphs

WHEN TO USE: the motion is given, or wanted, as a graph rather than a formula.

Graph	slope gives	area gives
x–t	velocity v_x	(not used)
v_x–t	acceleration a_x	displacement Δx *(Contact 2)*
a_x–t	rate of change of a_x	change in velocity Δv_x *(Contact 2)*

PROCEDURE: **Translate both ways:** graph $\rightarrow$ description of the motion, and description $\rightarrow$ sketched graph. Read a slope as the vertical change divided by the horizontal change, with units.

WATCH OUT: the graph is not the path; an upward graph does not mean the object moves upward.

CHECK: does each segment's sign and steepness agree with your motion description?

Contact 1: what we can now answer

Where is it? $\rightarrow$ position $\vec{r}(t)$

How has its position changed? $\rightarrow$ displacement $\Delta\vec{r}$

How fast and in what direction is its position changing? $\rightarrow$ velocity $\vec{v} = \mathrm{d}\vec{r}/\mathrm{d}t$

Connect: Lectures 3 and 4

The slope and area of motion graphs return in Lecture 4, where the area under a force–position graph is the work done. Lecture 3 relates the acceleration you meet next to the force on the object.

But what if… the runner *turns round*? The velocity changes from $+5.0$ to $-4.0\,\mathrm{m/s}$. Suppose the turn takes $2.0\,\mathrm{s}$: how rapidly is the velocity changing, and what does the sign of that rate mean? $\rightarrow$ **Contact 2: acceleration**.

Equation anatomy: acceleration

EQUATION – average acceleration

$$a_{\text{av},x} = \frac{\Delta v_x}{\Delta t} = \frac{v_{2x} - v_{1x}}{t_2 - t_1}$$

Meaning. the change of velocity per unit time over an interval.

Variables. Δv_x is final minus initial velocity; Δt the elapsed time.

Units. $\mathrm{m/s^2}$, that is, $(\mathrm{m/s})$ per second.

Interpretation. compare its sign with that of v_x: same sign, speeding up; opposite, slowing down.

Example. a car with v_x from $+10$ to $+30\,\mathrm{m/s}$ in $4.0\,\mathrm{s}$: $+5.0\,\mathrm{m/s^2}$.

EQUATION – instantaneous

$$a_x = \frac{\mathrm{d}v_x}{\mathrm{d}t} = \frac{\mathrm{d}^2x}{\mathrm{d}t^2}$$

Meaning. the rate of change of velocity at one instant.

Variables. $v_x(t)$ is the velocity function; $x(t)$ the position function.

Units. $\mathrm{m/s^2}$.

Interpretation. the slope of the v_x–t graph; the curvature of the x–t graph.

Example. $x = 2t^2$: $v_x = 4t$, $a_x = 4\,\mathrm{m/s^2}$, constant.

TOOLKIT 3 – Reading motion graphs

WHEN TO USE: the motion is given, or wanted, as a graph. Now slope *and* area are used.

Graph	slope gives	area gives
x–t	velocity v_x	(not used)
v_x–t	acceleration a_x	displacement Δx
a_x–t	rate of change of a_x	change in velocity Δv_x

PROCEDURE: Areas above the axis add; areas below subtract from the displacement but still add to the distance.

WATCH OUT: where $v_x = 0$ the object may reverse, so split the distance calculation there.

CHECK: do the signs and units of the answer match the graph?

But what if... the graph is a smooth curve $a_x(t)$? The area becomes an **integral**.

TOOLKIT 2 – The calculus of motion (completed)

WHEN TO USE: the acceleration (or velocity) is given as a function of time and you need the velocity and position.

PROCEDURE: **Integrate** from acceleration back to position:

$$v_x(t) = v_{0x} + \int_0^t a_x \, \mathrm{d}t' \qquad x(t) = x_0 + \int_0^t v_x \, \mathrm{d}t'$$

Add the initial conditions v_{0x} and x_0; without them the integral is defined only up to a constant. These integrals are the areas under a_x–t (giving Δv_x) and v_x–t (giving Δx), Toolkit 3.

WATCH OUT: do not forget the initial value.

CHECK: $v(0) = v_0$ and $x(0) = x_0$; differentiate your answer to recover the $a(t)$ you started with.

EQUATION – equation (1)

$$v_x = v_{0x} + a_x t$$

Meaning. the velocity changes by the same amount every second.

Variables. v_{0x} initial velocity; a_x constant acceleration; t elapsed time.

Units. $\mathrm{m/s} = \mathrm{m/s} + (\mathrm{m/s^2})(\mathrm{s})$.

Interpretation. a straight line of slope a_x on the v_x–t graph.

Example. the braking example: $t = (0 - 25.0)/(-6.0) = 4.2\,\mathrm{s}$.

EQUATION – equation (3)

$$v_x^2 = v_{0x}^2 + 2a_x(x - x_0)$$

Meaning. links the speeds to the displacement, with no time.

Variables. $x - x_0$ is the displacement; the others as in (1).

Units. $\mathrm{m^2/s^2} = \mathrm{m^2/s^2} + (\mathrm{m/s^2})(\mathrm{m})$.

Interpretation. the change in v_x^2 is proportional to the displacement.

Example. the braking example: $x - x_0 = 52\,\mathrm{m}$.

TOOLKIT 4 – Choosing a constant-acceleration equation

WHEN TO USE: the acceleration is constant and the motion is along a straight line, with some of $x - x_0$, v_{0x}, v_x, a_x, t known and one or two wanted.

PROCEDURE: **1.** Choose the origin and positive direction (Toolkit 1). **2.** List the known quantities. **3.** Identify the target. **4.** Find the variable that is *neither known nor wanted*. **5.** Choose the equation that omits it. **6.** Solve algebraically first, then substitute with units.

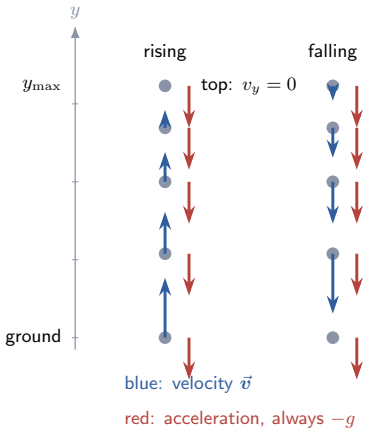
missing	$x - x_0$	v_x	t	a_x	v_{0x}
use	(1)	(2)	(3)	(4)	(5)

WATCH OUT: matching formulas to symbols without checking that a_x is constant, or that the motion is on a line.

CHECK: sign, units and physical meaning; does the answer agree with your prediction and the sketch?

A ball thrown upward: the numbers

PRACTICE



A ball is thrown upward at $14.7\,\mathrm{m/s}$ from the ground ($+y$ up).

Top: $v_y = 0 \Rightarrow t = \dfrac{14.7}{9.8} = 1.50\,\mathrm{s}$

$$y_{\max} = \frac{14.7^2}{2(9.8)} = 11.0\,\mathrm{m}$$

Back at ground: $t = 3.00\,\mathrm{s}$ with $v_y = -14.7\,\mathrm{m/s}$.

Neglecting air resistance, and returning to the same height: time up = time down, and the speed is the same at the same height on the way up and down.

But what if... the ball is thrown from the edge of a building, so it lands *below* its launch point? The equation for the landing time is then a quadratic with two roots. Which one is physical?

Your turn: a stone from a bridge

Independent problem

A stone is dropped from rest from a bridge and hits the water $2.50\,\mathrm{s}$ later. How high is the bridge, and how fast is the stone moving when it hits?

Predict first

Sketch with $+y$ up. Sign of v_y at impact? About how far does it fall in the first second, in the second second?

Independent problem

A stone is dropped from rest from a bridge and hits the water $2.50\,\mathrm{s}$ later. How high is the bridge, and how fast is the stone moving when it hits?

Predict first

Sketch with $+y$ up. Sign of v_y at impact? About how far does it fall in the first second, in the second second?

$y - y_0 = v_{0y}t + \frac{1}{2}a_y t^2 = 0 + \frac{1}{2}(-9.80)(2.50)^2 = -30.6\,\mathrm{m}$, so the bridge is $30.6\,\mathrm{m}$ high.
$v_y = 0 + (-9.80)(2.50) = -24.5\,\mathrm{m/s}$ (downward).
Evaluate: negative displacement and velocity, as predicted; check (3):
$v_y^2 = 2(9.8)(30.6) = 600$, $|v_y| = 24.5\,\mathrm{m/s}$ ✓

Connect: where Contact 2 is used next

Connect: Lectures 3 to 5

$a_x = F_x/m$ in Lecture 3; $v_x^2 = v_{0x}^2 + 2a_x(x - x_0)$ becomes the work–energy theorem in Lecture 4; free fall leads to gravitational potential energy in Lecture 5.

Physical meaning

Slopes and areas of graphs, integration with initial conditions, and the constant-acceleration equations with their conditions are the core of one-dimensional kinematics for the rest of the course.

Contact 2: what we can now do

Position
⇓ derivative
Velocity
⇓ derivative
Acceleration

and back again by integration with initial conditions.

You should be able to

- decide from the signs of v_x and a_x whether the speed increases;
- translate graphs, and integrate or differentiate;
- choose a constant-acceleration equation and say when it applies;
- solve free fall with a consistent sign convention and the physical root.

But what if… the motion is not on a line? A ball kicked at 30° moves both ways at once; a car on a bend has constant speed but $\vec{a} \neq 0$; a walker on a train has two velocities. → **Contact 3**.

Equation anatomy: projectile motion

EQUATION – horizontal motion

$$v_x = v_0 \cos\theta$$

$$x = x_0 + v_0 \cos\theta \, t$$

Meaning. no horizontal acceleration, so the horizontal velocity is constant.

Variables. v_0 launch speed; θ launch angle above the horizontal; t time.

Units. $\mathrm{m/s}$ and m.

Interpretation. the range is this constant speed multiplied by the time of flight.

Example. a long hit: $23.0\,\mathrm{m/s} \times 3.94\,\mathrm{s} = 90.4\,\mathrm{m}$ (unrounded values).

EQUATION – vertical motion

$$v_y = v_0 \sin\theta - gt$$

$$y = y_0 + v_0 \sin\theta \, t - \tfrac{1}{2} g t^2$$

Meaning. constant downward acceleration g acting on the vertical motion alone.

Variables. y_0 launch height; $g = 9.80\,\mathrm{m/s^2}$; $+y$ upward.

Units. $\mathrm{m/s}$ and m.

Interpretation. a parabola in t; $v_y = 0$ only at the top.

Example. a long hit: top at $t = 19.3/9.80 = 1.97\,\mathrm{s}$.

TOOLKIT 5 – Projectile motion

WHEN TO USE: an object moves freely under gravity alone (air resistance neglected) and its launch velocity is not along the vertical.

PROCEDURE: **1. Sketch** the trajectory and mark the start, top and end. **2. Axes:** $+x$ horizontal, $+y$ up, origin at launch (Toolkit 1). **3. Resolve** $\vec{v}_0$ into v_{0x} and v_{0y} (Lecture 1, Toolkit 3). **4.** x**-motion:** $a_x = 0$. **5.** y**-motion:** $a_y = -g$ (Toolkit 4 for the equation). **6. Same time** in both directions; it is the link. **7. Reconstruct** the speed and direction if needed (Lecture 1, Toolkit 4).

WATCH OUT: roots of a quadratic in t: choose the physical one; $v_y = 0$ only at the top, not at landing.
CHECK: does the result agree with the sketch, the prediction and the signs of v_x and v_y?

But what if... we try it on a real launch from level ground? Predict first, then calculate.

Independent problem

A ball is kicked at $20.0\,\mathrm{m/s}$, 30.0° above the horizontal, from the edge of a $10.0\,\mathrm{m}$ high roof. Find the time to reach the ground, the horizontal distance from the roof, and the velocity (magnitude and direction) at impact.

Predict first

Sketch. Is the time more or less than $2(10.0)/9.80 = 2.04\,\mathrm{s}$? Is the impact velocity above or below the horizontal?

Launched from a height: solution

PRACTICE

Set up: $v_{0x} = 17.3\,\text{m/s}$, $v_{0y} = 10.0\,\text{m/s}$, $y_0 = 10.0\,\text{m}$, $y = 0$.

Launched from a height: solution

Set up: $v_{0x} = 17.3\,\text{m/s}$, $v_{0y} = 10.0\,\text{m/s}$, $y_0 = 10.0\,\text{m}$, $y = 0$.

Execute: $0 = 10.0 + 10.0t - 4.9t^2 \Rightarrow t = \dfrac{10.0 \pm \sqrt{100 + 196}}{9.8} = 2.78\,\text{s}$ or $-0.74\,\text{s}$.

Range: $17.3 \times 2.78 = 48.1\,\text{m}$.

$v_y = 10.0 - 9.8(2.78) = -17.2\,\text{m/s}$; $v = \sqrt{17.3^2 + 17.2^2} = 24.4\,\text{m/s}$,

$\theta = \tan^{-1}\dfrac{17.2}{17.3} = 44.8°$ below the horizontal.

Launched from a height: solution

Set up: $v_{0x} = 17.3\,\text{m/s}$, $v_{0y} = 10.0\,\text{m/s}$, $y_0 = 10.0\,\text{m}$, $y = 0$.

Execute: $0 = 10.0 + 10.0t - 4.9t^2 \Rightarrow t = \dfrac{10.0 \pm \sqrt{100 + 196}}{9.8} = 2.78\,\text{s}$ or $-0.74\,\text{s}$.

Range: $17.3 \times 2.78 = 48.1\,\text{m}$.

$v_y = 10.0 - 9.8(2.78) = -17.2\,\text{m/s}$; $v = \sqrt{17.3^2 + 17.2^2} = 24.4\,\text{m/s}$,

$\theta = \tan^{-1}\dfrac{17.2}{17.3} = 44.8°$ below the horizontal.

Evaluate: $2.78\,\text{s} > 2.04\,\text{s}$ as predicted; $v_y < 0$; the landing speed exceeds $20.0\,\text{m/s}$ because the ball has fallen $10\,\text{m}$ below its launch. ✓

But what if... we want the *shape* of the path, not only the landing point? $x(t)$ and $y(t)$ together are the **parametric equations** of the trajectory.

Parametric equations: the trajectory (supporting tool)

$x(t)$ and $y(t)$ together are the **parametric equations** of the path; t is the parameter. Eliminating t gives the shape of the trajectory.

Launch point at $x_0 = y_0 = 0$

$$y = x\tan\theta - \frac{g}{2v_0^2\cos^2\theta}\,x^2$$

a parabola.

Example

$x = 3.0t,\ \ y = 4.0t - 4.9t^2$ (SI). Eliminate t and find where $y = 0$ again.

Parametric equations: the trajectory (supporting tool)

$x(t)$ and $y(t)$ together are the **parametric equations** of the path; t is the parameter. Eliminating t gives the shape of the trajectory.

Launch point at $x_0 = y_0 = 0$

$$y = x\tan\theta - \frac{g}{2v_0^2\cos^2\theta}\,x^2$$

a parabola.

Example

$x = 3.0t,\ \ y = 4.0t - 4.9t^2$ (SI). Eliminate t and find where $y = 0$ again.

$t = x/3.0$, so $y = 1.33x - 0.544x^2$.
$y = 0$: $x = 1.33/0.544 = 2.45\,\mathrm{m}$.

But what if... gravity changes both the size and the direction of $\vec{v}$? What if only the *direction* changes, at constant speed? $\rightarrow$ **circular motion**.

EQUATION – radial acceleration

$$a_{\text{rad}} = \frac{v^2}{R} = \omega^2 R$$

Meaning. the acceleration that turns the velocity around the circle.

Variables. v speed; R radius; ω angular speed.

Units. $\mathrm{m/s^2}$; ω in $\mathrm{rad/s}$.

Interpretation. directed toward the centre; doubling the speed quadruples it.

Example. the bend in the road: $20.0^2/80.0 = 5.0\,\mathrm{m/s^2}$.

EQUATION – T, f and ω

$$T = \frac{1}{f}, \qquad \omega = 2\pi f = \frac{2\pi}{T}, \qquad v = \omega R$$

Meaning. how often, and how fast in angle, the object goes round.

Variables. T period; f frequency; ω angular speed.

Units. T in s; f in Hz $= \mathrm{s^{-1}}$; ω in rad/s.

Interpretation. one revolution is 2π rad in time T.

Example. the centrifuge: $f = 50\,\mathrm{Hz}$, $\omega = 314\,\mathrm{rad/s}$.

TOOLKIT 6 – Circular motion

WHEN TO USE: an object moves on a circle (or a circular arc) and you need its acceleration, speed, period or frequency.

PROCEDURE: **1. Sketch** the circle and mark $\vec{v}$ (tangent) and $\vec{a}$ (to the centre) at one point. **2. Convert** to SI first: $f = \text{rev/min}/60$, $\omega = 2\pi f = 2\pi/T$. **3. Link** $v = \omega R = 2\pi R/T$. **4. Choose** $a_{\text{rad}} = v^2/R$ or $\omega^2 R$, whichever uses the given quantities.

RULE: constant speed does not imply zero acceleration.
WATCH OUT: ω is in rad/s, not rev/min or degrees.
CHECK: $v^2/R = \omega^2 R$ (the two forms agree); $\vec{a}$ points to the centre; its size is reasonable compared with g.

But what if... the car is also speeding up while it rounds the bend? Then $a_{\text{rad}} = v^2/R$ is only *part* of the acceleration.

Rule: general curved motion

$$a_t = \frac{\mathrm{d}v}{\mathrm{d}t} \quad \text{changes the speed}$$

$$a_n = \frac{v^2}{\rho} \quad \text{changes the direction}$$

ρ is the local radius of curvature of the path. For uniform circular motion $a_t = 0$ and $\rho = R$, so $a_n = a_{\text{rad}}$.

Link to Contact 2

Tangential acceleration is the component **parallel** to $\vec{v}$ ($a_\parallel$); normal acceleration is the component **perpendicular** to $\vec{v}$ ($a_\perp$).

Predict first

A car speeds up while rounding a bend. Which component is non-zero, a_t, a_n or both?

But what if... we put numbers in? Use the toolkit on a rotating sample, then on a car.

Independent problem

A car travels at a constant $20.0\,\mathrm{m/s}$ around a circular bend of radius $80.0\,\mathrm{m}$. Find its radial acceleration, the angular speed, and the time to complete a full circle (if the road were a circle).

Predict first

Draw the direction of $\vec{v}$ and $\vec{a}$ at one point on the circle. Is a larger or smaller than g?

Independent problem

A car travels at a constant $20.0\,\mathrm{m/s}$ around a circular bend of radius $80.0\,\mathrm{m}$. Find its radial acceleration, the angular speed, and the time to complete a full circle (if the road were a circle).

Predict first

Draw the direction of $\vec{v}$ and $\vec{a}$ at one point on the circle. Is a larger or smaller than g?

$a_{\text{rad}} = v^2/R = 400/80.0 = 5.0\,\mathrm{m/s^2}$ (toward the centre), about $0.5\,g$. $\omega = v/R = 0.25\,\mathrm{rad/s}$. $T = 2\pi R/v = 25.1\,\mathrm{s}$.

Evaluate: check $\omega^2 R = 0.0625 \times 80.0 = 5.0\,\mathrm{m/s^2}$ ✓

But what if... a passenger walks along the aisle of the car, or of a train, while it moves? Then we must ask *whose* velocity we mean. → **relative velocity**.

TOOLKIT 7 – Relative velocity

WHEN TO USE: a velocity is known relative to one observer and wanted relative to another (train and ground, boat and water, aircraft and air).

PROCEDURE: **1. Identify** the object P and the observers A and B. **2. Write the relation** $\vec{v}_{P/A} = \vec{v}_{P/B} + \vec{v}_{B/A}$ *before* inserting values. **3. Sketch** the vector triangle and choose axes (Toolkit 1). **4. In 2D,** use components (Lecture 1, Toolkits 3 and 5) and reconstruct (Lecture 1, Toolkit 4).

WATCH OUT: do not add speeds: add vectors.
CHECK: the inner subscripts cancel, and the direction agrees with the sketch.

Equation anatomy: relative velocity

EQUATION – relative velocity

$$\vec{\boldsymbol{v}}_{P/A} = \vec{\boldsymbol{v}}_{P/B} + \vec{\boldsymbol{v}}_{B/A}$$

Meaning. the velocity of P seen from A is its velocity seen from B plus the velocity of B seen from A.

Variables. P the moving object; A and B two observers or frames.

Units. $\mathrm{m/s}$ for every term.

Interpretation. velocities are vectors, so they add as vectors; the inner subscript B cancels.

Example. the passenger on the train, then the river crossing.

Reading the subscripts

$\vec{\boldsymbol{v}}_{P/T}$: passenger relative to train.

$\vec{\boldsymbol{v}}_{T/G}$: train relative to ground.

$\vec{\boldsymbol{v}}_{P/G} = \vec{\boldsymbol{v}}_{P/T} + \vec{\boldsymbol{v}}_{T/G}$.

Decide what physics applies; the method is not given

A stone on a string $0.60\,\mathrm{m}$ long is whirled at 2.5 revolutions per second in a horizontal circle, $1.40\,\mathrm{m}$ above level ground. The string breaks while the stone is moving horizontally.

(a) What is the stone's radial acceleration just before the string breaks?
(b) How far does the stone travel horizontally before it lands, and what is its speed on landing? (Neglect air resistance.)

Predict first

IDENTIFY $\rightarrow$ SET UP $\rightarrow$ PREDICT $\rightarrow$ EXECUTE $\rightarrow$ EVALUATE. Which model describes each stage? Is the radial acceleration much larger or much smaller than g?

Identify: stage 1 is uniform circular motion (Toolkit 6); stage 2 is a projectile launched horizontally from $1.40\,\mathrm{m}$ (Toolkit 5).
Set up: $f = 2.5\,\mathrm{Hz}$, $R = 0.60\,\mathrm{m}$. At the break, $\vec{\boldsymbol{v}}$ is tangent and horizontal, so $v_{0x} = v$ and $v_{0y} = 0$.

Identify: stage 1 is uniform circular motion (Toolkit 6); stage 2 is a projectile launched horizontally from $1.40\,\mathrm{m}$ (Toolkit 5).

Set up: $f = 2.5\,\mathrm{Hz}$, $R = 0.60\,\mathrm{m}$. At the break, $\vec{\boldsymbol{v}}$ is tangent and horizontal, so $v_{0x} = v$ and $v_{0y} = 0$.

Execute (a): $\omega = 2\pi f = 15.7\,\mathrm{rad/s}$, $v = \omega R = 9.42\,\mathrm{m/s}$,

$$a_{\mathrm{rad}} = \frac{v^2}{R} = \frac{88.8}{0.60} = 148\,\mathrm{m/s^2} \approx 15\,g.$$

Your turn: an integrated problem, solution

Identify: stage 1 is uniform circular motion (Toolkit 6); stage 2 is a projectile launched horizontally from $1.40\,\mathrm{m}$ (Toolkit 5).

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$$a_{\text{rad}} = \frac{v^2}{R} = \frac{88.8}{0.60} = 148\,\mathrm{m/s^2} \approx 15\,g.$$

(b): fall time $t = \sqrt{2(1.40)/9.80} = 0.535\,\mathrm{s}$; horizontal distance $9.42 \times 0.535 = 5.04\,\mathrm{m}$.
$v_y = -9.80(0.535) = -5.24\,\mathrm{m/s}$, so $v = \sqrt{9.42^2 + 5.24^2} = 10.8\,\mathrm{m/s}$, at $\tan^{-1}(5.24/9.42) = 29.1^\circ$ below the horizontal.

Evaluate: much larger than g, as predicted; the landing speed exceeds the break-off speed because the stone has fallen; check $\sqrt{2gh + v^2} = \sqrt{27.4 + 88.8} = 10.8\,\mathrm{m/s}$ ✓

Connect: where Contact 3 is used next

Projectiles

Lecture 5 finds the speed at any height by energy conservation, with no time; Lecture 7 treats collisions in two dimensions with the same component method.

Circular motion

Lecture 3 supplies the force behind $a_{\text{rad}} = v^2/R$ (centripetal force); Lecture 8 extends ω, T and f to rotating bodies.

Relative velocity

Lecture 7: a collision looks different in different frames, and the velocity of the centre of mass is the natural frame in which to describe it.

Physical meaning

Independent components, the constant-acceleration equations and $\vec{\boldsymbol{v}} = \mathrm{d}\vec{\boldsymbol{r}}/\mathrm{d}t$ remain the core of the motion toolbox for the rest of the course.

What you should remember: the chain, Contact 1 (part 1)

Concept	Limitation that motivates the next	Leads to
Opening journey	"how far?" has two answers: $60\,\mathrm{m}$ from the start, $140\,\mathrm{m}$ of road	**Vocabulary**
Vocabulary	the same word names a vector in one place and a number in another	**Notation**
Notation	the sign of v_x depends on which way is positive	**Position, displacement**
Position, displacement	$\Delta x = +60\,\mathrm{m}$ but distance $140\,\mathrm{m}$; reversing the axis flips every sign	**Toolkit 1**
Toolkit 1	how rapidly did the position change over $30\,\mathrm{s}$?	**Average velocity**
Average velocity, dragster	$+2.0\,\mathrm{m/s}$ hides $+5.0$ then $-4.0\,\mathrm{m/s}$; the dragster has no single speed	**Instantaneous velocity**

What you should remember: the chain, Contact 1 (part 2)

Concept	Limitation that motivates the next	Leads to
Instantaneous velocity	the chord table only approaches $12\,\mathrm{m/s}$; we need it exactly	**Derivative, anatomy**
Velocity anatomy, concept check	the motion is not along one line	**Vectors**
Vectors	$x(t)$, $y(t)$, $z(t)$ given as formulas: how do we get $\vec{v}$?	**Toolkit 2**
Toolkit 2, helix example	only a graph, no formula	**Toolkit 3**
Toolkit 3	velocity itself changes: $+5.0$ to $-4.0\,\mathrm{m/s}$ in $2.0\,\mathrm{s}$	**Acceleration (Contact 2)**

What you should remember: the chain, Contact 2 (part 1)

Concept	Limitation that motivates the next	Leads to
Opening question	we need the rate of change of velocity itself	**Average acceleration**
Average acceleration	the runner's $-4.5\,\mathrm{m/s^2}$ is an average over $2.0\,\mathrm{s}$	**Instantaneous acceleration**
Instantaneous acceleration	$a_x < 0$ throughout the turn, yet the runner first slows, then speeds up	**Sign rule**
Sign rule	can we see it with no numbers?	**Motion diagrams**
Motion diagram	the velocity changes direction as well as size	**Acceleration along, across**
Acceleration along, across	a v–t graph of straight segments: slopes give a_x, but displacement?	**Toolkit 3 completed**
Toolkit 3 completed	the graph is a smooth curve given by a formula	**Integration**

What you should remember: the chain, Contact 2 (part 2)

Concept	Limitation that motivates the next	Leads to
Integration, Toolkit 2 completed	integrating every time is slow when a_x is constant	**Constant-acceleration equations**
Constant-acceleration equations	they fail if we use them with $a_x = 6.0t - 4.0$: 39 against $12\,\mathrm{m/s}$	**Counter-example**
Counter-example	with a_x constant, five equations: which one?	**Toolkit 4**
Toolkit 4, braking distance	every object near the ground has the same a_y	**Free fall**
Free fall, ball thrown upward	landing below the launch point gives two roots	**Two roots**
Two roots, stone	the motion is not on one line	**Contact 3**

What you should remember: the chain, Contact 3 (part 1)

Concept	Limitation that motivates the next	Leads to
Monkey and dart	how do horizontal and vertical motions combine?	**Independent components**
Independent components	height, time of flight and range as numbers	**Projectile model, Toolkit 5**
Model, anatomy, Toolkit 5	a real launch from level ground	**Long hit**
Long hit	landing below the launch height: not symmetric	**Launched from a height**
Launched from a height	the shape of the whole path	**Parametric equations**
Parametric equations	only the direction of $\vec{v}$ changes, at constant speed	**Circular motion**
Circular geometry	how large is $\vec{a}$?	$a_{\text{rad}} = v^2/R$

What you should remember: the chain, Contact 3 (part 2)

Concept	Limitation that motivates the next	Leads to
$a_{\text{rad}} = v^2/R$	the data are revolutions per second, not v	**Period, frequency, ω**
Period, frequency, Toolkit 6	the car also speeds up on the bend	**Tangential and normal acceleration**
Tangential, normal	put numbers in	**Centrifuge, bend**
Bend in the road	a passenger walks inside the moving car: whose velocity?	**Relative velocity**
Relative velocity, Toolkit 7	the velocities are not along one line	**River crossing**
River crossing	several ideas at once: which model, and when?	**Integrated applications**

THE SEVEN TOOLKITS

1	**Signs and coordinates**	origin and positive direction first; signs relative to your axis; final minus initial
2	**Calculus of motion**	differentiate $\vec{r} \to \vec{v} \to \vec{a}$; integrate back with initial conditions
3	**Reading motion graphs**	slope and area for x–t, v–t, a–t; translate both ways
4	**Choosing a constant-a equation**	list known, wanted, missing; confirm a constant
5	**Projectile motion**	resolve; independent x and y; same time; physical root
6	**Circular motion**	$\vec{v}$ tangent, $\vec{a}$ to the centre; $a_{\text{rad}} = v^2/R = \omega^2 R$
7	**Relative velocity**	subscript chain; vectors, not speeds

The master procedure sits above these. Lecture 1 toolkits (components, reconstructing a vector, adding by components) are reused, not renamed.